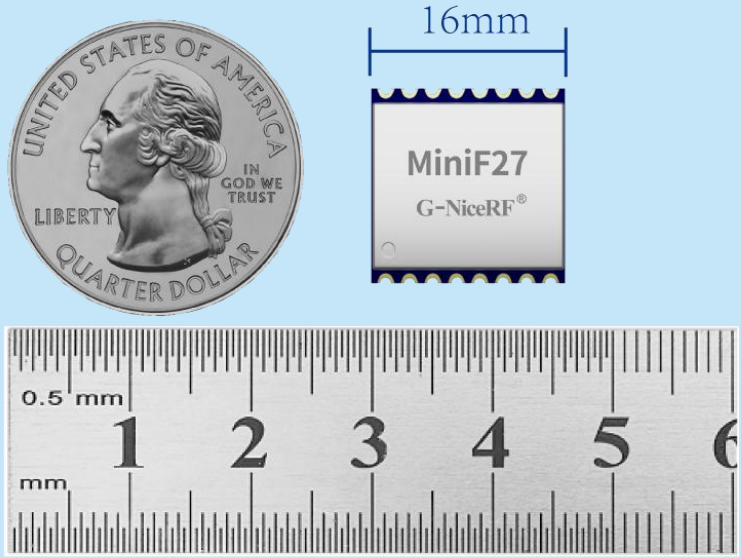
MiniF27 Wireless transceiver module

- Ultra-compact size, same as LoRa low-Power modules
- High power 800mW @ 5V

Product Specification



Catalogue

Note: Revision History

Revision	Date	Comment
V1.0	2025-3	First release

1. Descriptions

MiniF27 wireless module is designed using Semtech's LoRa chip, featuring a high-precision crystal oscillator, ultra-low receiving and sleep currents, and a sensitivity of -148dBm. It can wake up the microcontroller periodically while maintaining low power consumption. The module's antenna switch is internally controlled by the chip, reducing the resource demand on the external MCU. Despite its ultra-compact size (same as the LoRa1262), the MiniF27 delivers an output power of up to 27dBm (500mW). Its high integration, small size, and high power make the MiniF27 highly advantageous in IoT and battery-powered applications.

MiniF27 strictly uses lead-free technology for production and testing, and meets RoHS and Reach standards.

Module Type	Built-in chip	Operating rate	Sensitivity
1262MiniF27	SX1262	0.018-62.5Kpbs	-148dBm@BW=10.4KHz,SF=12
CC68MiniF27	LLCC68	1.76-62.5Kpbs	-129dBm@BW=125KHz,SF=9

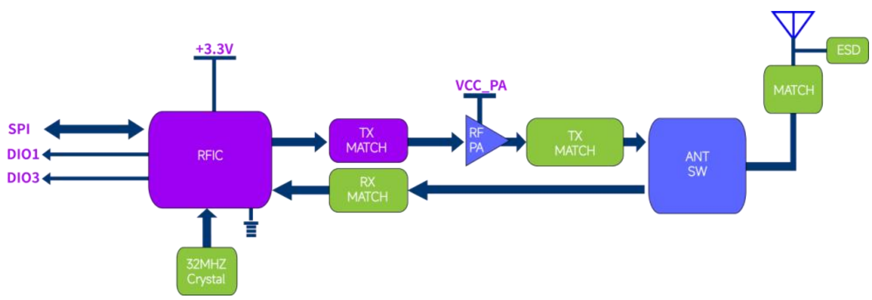
2. Features

- Frequency Range:433/470 MHz （customizable150-960 MHz）
- Sensitivity: -148dBm @LoRa SX1262
- 256 bytes FiFo
- Maximum output power：29 dBm（800mW）
- LoRa, (G)FSK modulation
- Data transfer rate: 0.6-300 Kbps @FSK

3. Applications

- Industrial meter reading
- Parking lot sensor management
- Industrial automation
- Agricultural sensor
- Smart city
- Remote control
- Street lights
- Logistics management
- Environmental sensor
- Health products
- Security products
- Warehouse management

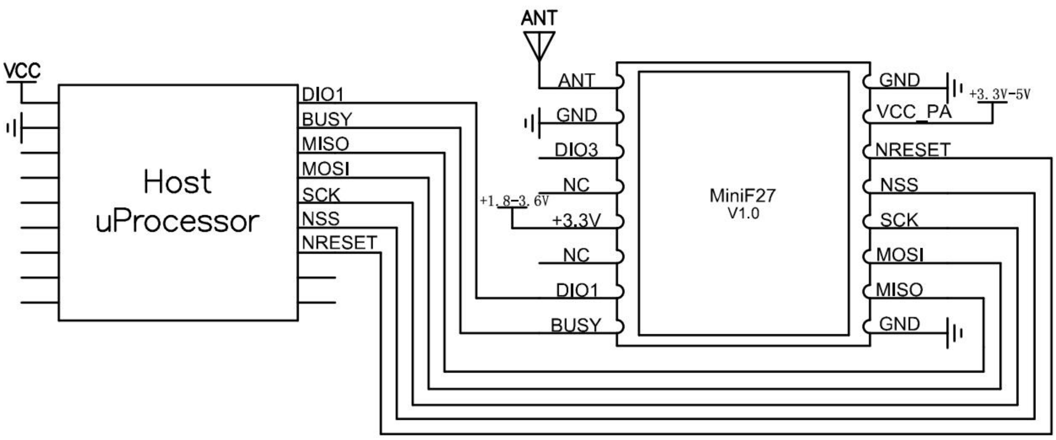
4. Block Diagram



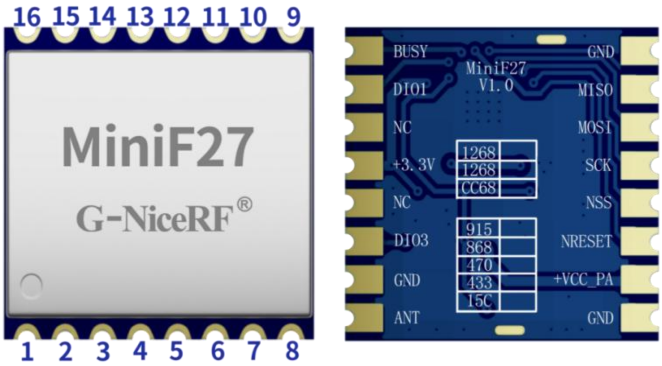
5. Electrical Characteristics

Parameter	Min	Typ	Max	Unit	Condition
Operation Condition					
Working voltage	1.8	3.3	3.6	V	Module + 3.3V pin
	3.3	4.0	5.0	V	Module + VCC_PA pin
Temperature range	-40		85	℃	
Current Consumption					
RX current		6		mA	@DCDC Mode
TX current		500		mA	@5v,29dBm
Sleep current		1		uA	
RF Parameter					
Frequency range	410	433	450	MHz	@433MHz
	470	490	510	MHz	@470MHz
Modulation Rate	0.018		62.5	kbps	@SX1262
	1.76		62.5	kbps	@LLCC68
	0.6		300	kbps	@FSK
Output power	0		27	dBm	
Receiving sensitivity		-148		dBm	@SX1262, BW=10.4KHz SF =12
		-129		dBm	@LLCC68, BW=125KHz SF =9

6. Typical application circuit



7. Pin definition



Pin NO	Pin name	IO Level	Description
1,8,10	GND		power ground
2	MISO	0-3.3V	SPI Output for SPI data
3	MOSI	0-3.3V	SPI Input for SPI data
4	SCK	0-3.3V	SPI Clock Input
5	NSS	0-3.3V	SPI Chip Select Input
6	NRESET	0-3.3V	Reset input
7	+VCC_PA		power supply pin connected to the internal amplifier. The higher the voltage, the higher the power. Voltage range: 3.3-5V
9	ANT		Connect to a 50Ω coaxial antenna
11	DIO3	0-3.3V	It can be customized as an interrupt signal indicator. For details, refer to the chip documentation.
12,14	NC		
13	+3.3V		Chip power supply pin, voltage range: 1.8-3.6V
15	DIO1	0-3.3V	Can be customized as interrupt signal indication, see chip data for details
16	BUSY	0-3.3V	Used for status indication, see datasheet for details.

8. Performance Parameters

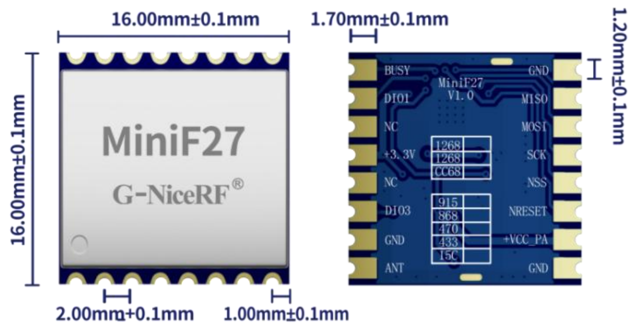
Output Power Relationship Diagram (dBm) @ 4V

Chip Output Level	Module Output Power (dBm)	Operating Current (mA)
0	9.8	110
1	12.8	120
2	14.8	130
3	17.5	155
4	20.9	200
5	22.9	250
6	25.7	330
7	27.1	400
8	28.2	440
9	28.7	460

Output Power vs. Voltage Chart (dBm)

VCC_PA Voltage (V)	Module Output Power (dBm)	Operating Current (mA)
3.3	26.6	364
3.6	27.4	385
4.0	28.1	424
4.3	28.6	453
4.6	29.3	473
5.0	29.8	504

9. Mechanism Dimension(Unit:mm)



Module Thickness: 2.1mm

10. Product order information

Product Name	Description
1262MiniF27-433	SX1262 chip， Working frequency 433MHz
1262MiniF27-470	SX1262 chip， Working frequency 470MHz
CC68MiniF27-433	LLCC68 chip， Working frequency 433MHz
CC68MiniF27-470	LLCC68 chip， Working frequency 470MHz

11. FAQ

a) Why module can not communicate properly？

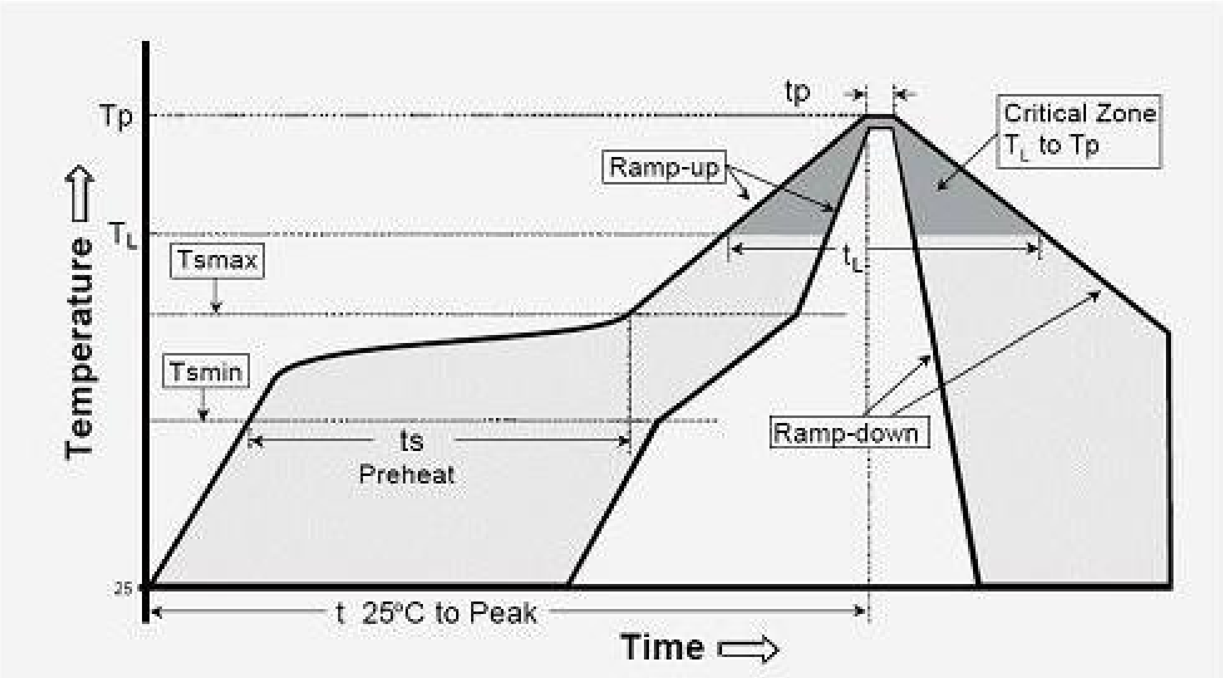
1) Check if there is power connection error；
2) Check if Module is in normal communication mode；
3) Check if frequency, channel, and mute are same；
4) Check if module is damaged；

b) Why transmission distance is not far as it should be？

1) Power supply ripple is too large；
2) The antenna types do not match, or not installed properly；
3) The same frequency interference；
4) The surrounding environment is harsh, strong interference sources.

Appendix : SMD Reflow Chart

Below reflow profile is recommended for SMT technology:



IPC/JEDEC J-STD-020B the condition for lead-free reflow soldering	big size components (thickness >=2.5mm)
The ramp-up rate (Tl to Tp)	3℃/s (max.)
preheat temperature - Temperature minimum (Tsmin) - Temperature maximum (Tsmax) - preheat time (ts)	 150℃ 200℃ 60~180s
Average ramp-up rate(Tsmax to Tp)	3℃/s (Max.)
- Liquidous temperature(TL)	217℃
- Time at liquidous(tL)	60~150 second
peak temperature(Tp)	245+/-5℃